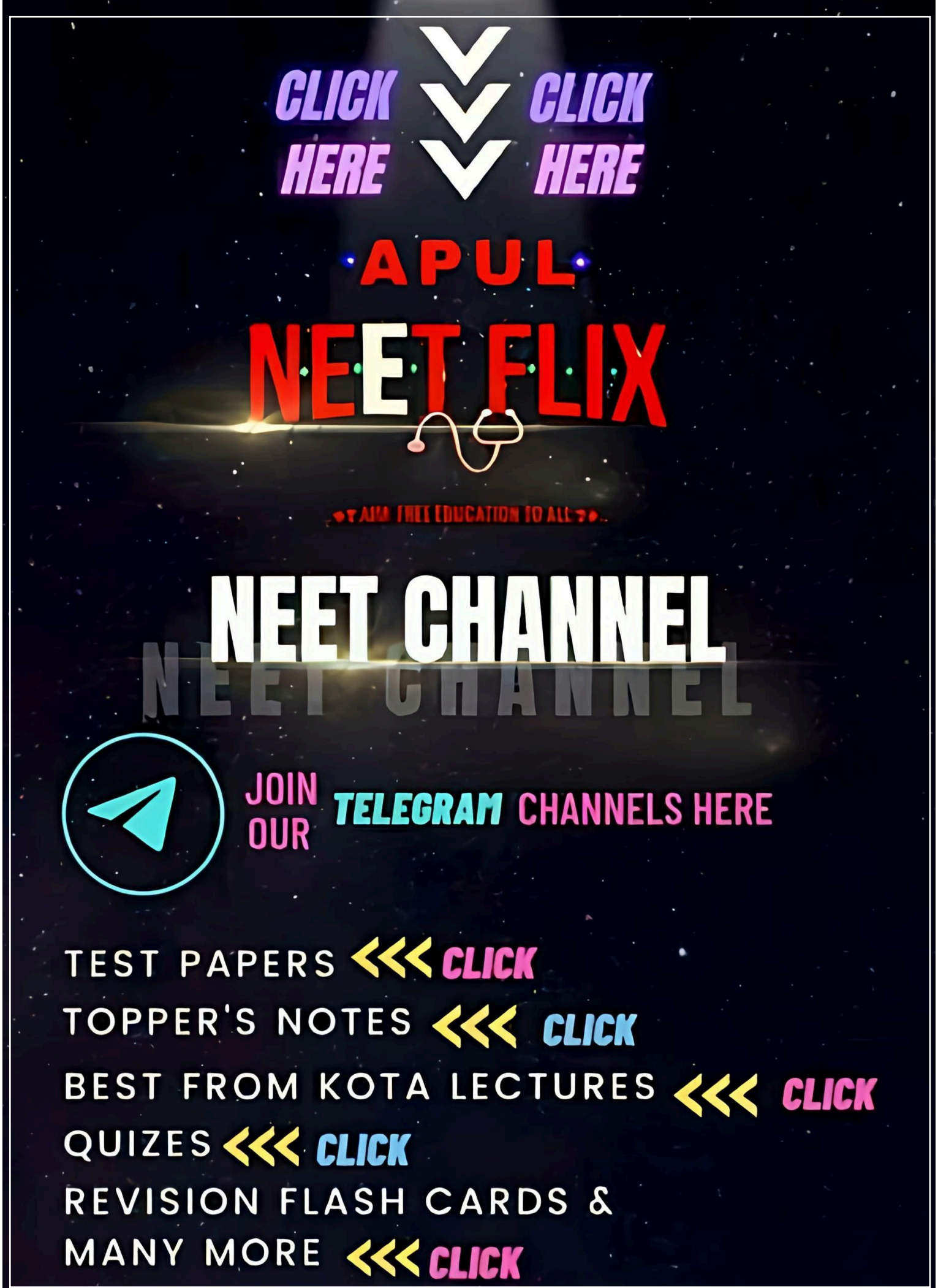

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PHYSICS

1) In meter bridge two unknown resistances A and B are connected in left and right gaps respectively. Null point is obtained when the bridge wire gets divided in ratio of 3 : 4. When both A and B are increased by 30 Ω, the null point divides the wire in ratio 6 : 7. The values of resistance A and B are respectively :

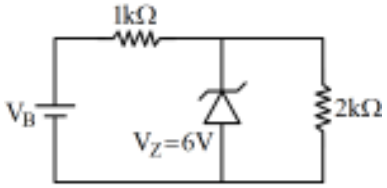
(1) 12Ω, 16Ω

(2) 30Ω, 40Ω

(3) 36Ω, 48Ω

(4) 45Ω, 60Ω

2) The given circuit represents the application of Zener diode as voltage regulator. The breakdown voltage for Zener diode is 6V. If the battery voltage V_B varies from 8V to 15V, the minimum and



maximum values of current through zener diode are respectively :

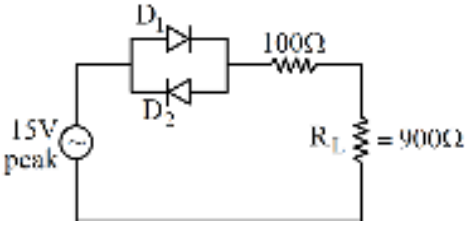
(1) 0.5 mA and 6 mA

(2) 1 mA and 6 mA

(3) 1.5 mA and 5 mA

(4) 0 mA and 6 mA

3) Two diodes D_1 and D_2 with barrier voltages 0.3V and 0.7V respectively are connected to a supply



of 15V peak AC along with two resistors : Match the terms in list-I with their suitable options in list-II.

List-I		List-II	
A.	Peak current in diode D_1 (in mA)	P.	12.9
B.	Peak voltage across load R_L when D_1 is forward biased (in volts)	Q.	14.7
C.	Peak voltage across R_L when D_2 is forward biased (in volts)	R.	1.47
D.	Maximum potential drop across 100 Ω resistor (in volts)	S.	13.2

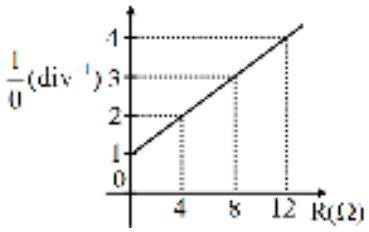
(1) A→S, B→Q, C→P, D→R

(2) A→P, B→Q, C→R, D→S

(3) A→Q, B→S, C→P, D→R

(4) A→Q, B→S, C→R, D→P

4) In an experiment to determine the resistance of a galvanometer by method of half deflection, the plot of $\left(\frac{1}{\theta}\right)$ vs the resistance R of a resistance box is shown in the figure. If the source emf is 2.5V, the figure of merit of galvanometer is given by :-



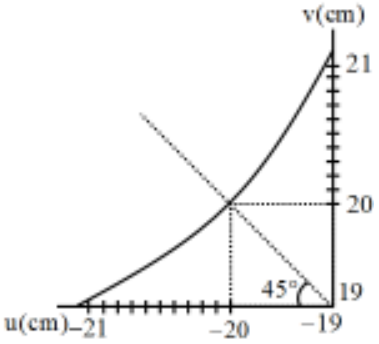
(1) 2.5×10^{-1} A/div

(2) 3×10^{-1} A/div

(3) 4.5×10^{-1} A/div

(4) 6.25×10^{-1} A/div

5) In an experiment to determine focal length of a lens by parallax method, a student obtained the graph between object position (u) and corresponding image position (v) w.r.t. lens as shown in figure. The focal length of lens can be estimated as :



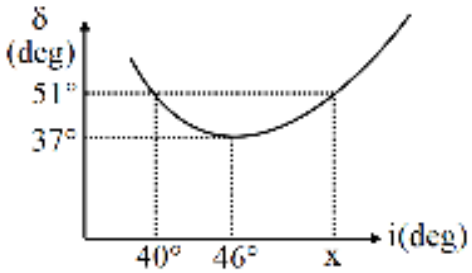
(1) 10cm ± 0.1cm

(2) 10cm ± 0.5cm

(3) 20cm ± 0.1cm

(4) 1.0cm ± 0.25cm

6) The given graph shows the variation of angle of deviation versus change in angle of incidence for a triangular prism kept in air. The value of angle of incidence x is :



(1) 56°

(2) 60°

(3) 66°

(4) 72°

7) In physics lab, a student is provided with four unlabelled components each exhibiting some different current – voltage (I–V) characteristics. Match the I–V characteristic in column-I with suitable component listed in column-II.

	Column-I		**Column-II**
A.	Straight line passing through origin	P.	Capacitor
B.	Non-linear curve with sharp knee at 0.7 V, conducts in one direction	Q.	LED
C.	Non-linear curve with sharp knee at 1.8V, conducts in one direction with glow	R.	Diode
D.	Does not conduct dc. But in a.c. current leads voltage by 90°	S.	Resistor

(1) A-S, B-R, C-Q, D-P

(2) A-S, B-P, C-Q, D-R

(3) A-S, B-Q, C-R, D-P

(4) A-Q, B-R, C-P, D-S

8) A dipole of 2×10^{-8} Cm is placed in a uniform electric field of 3×10^{5} N/C. The maximum torque on the dipole is :-

(1) 6×10^{-3} Nm

(2) 6×10^{-5} Nm

(3) 6×10^{-4} Nm

(4) 6×10^{-6} Nm

9) A metal sphere of radius R is given charge Q. Another uncharged metal sphere of radius 2R is brought in contact with it. The ratio of final charges on them is :-

(1) 1 : 2

(2) 2 : 1

(3) 1 : 4

(4) 4 : 1

10) A charge Q is divided into two parts and kept at a fixed distance r. The maximum possible force between the two is :-

(1) $\frac{KQ^2}{4r^2}$

(2) $\frac{KQ^2}{r^2}$

(3) $\frac{KQ^2}{2r^2}$

(4) $\frac{2KQ^2}{r^2}$

11) An isolated sphere of radius R has a potential of 200 V at its surface. The potential at a point 2R from centre is :-

(1) 200 V

(2) 100 V

(3) 50 V

(4) 400 V

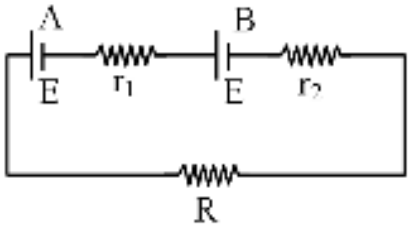
12) A 10 Ω resistor is connected across a 10 V battery for 10 minutes. The heat produced is :-

(1) 60 J

(2) 600 J

(3) 6000 J

(4) 1000 J



13) If terminal voltage of cell A is 0, the R is equal to :-

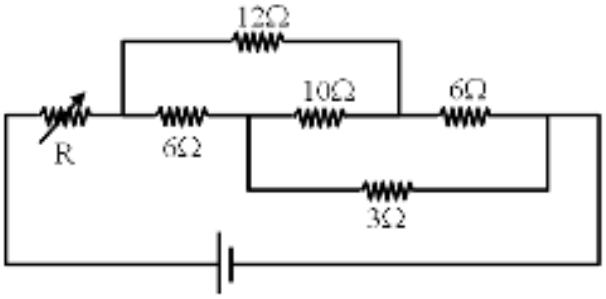
(1) $2r_2 - r_1$

(2) $2r_1 - r_2$

(3) $r_2 - r_1$

(4) $r_1 - r_2$

14) The value of resistance R, for which power in it is maximum, is :-



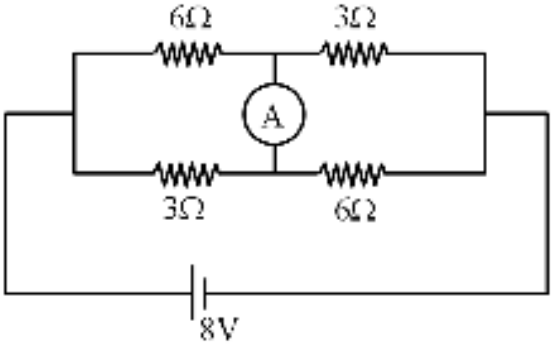
(1) 6 Ω

(2) 2 Ω

(3) 4/3 Ω

(4) 22 Ω

15) The reading of ammeter is :-



(1) $\frac{1}{3}$ A

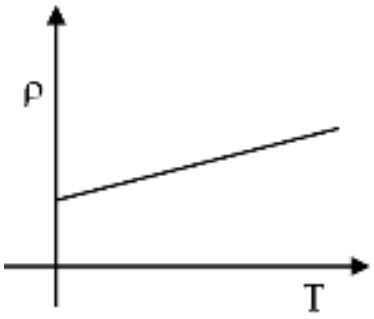
(2) $\frac{2}{3}$ A

(3) 1 A

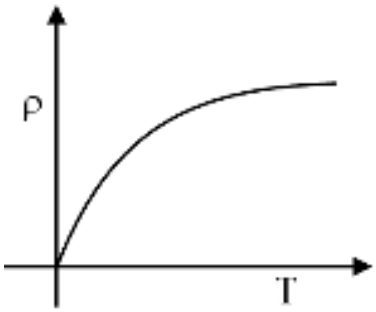
(4) 0

16) Which of the following graph is correct for semiconductors ?

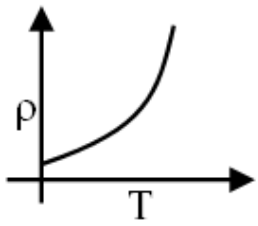
(1)



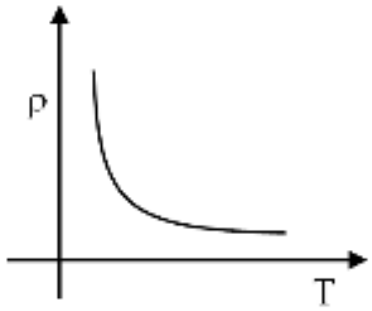
(2)



(3)



(4)



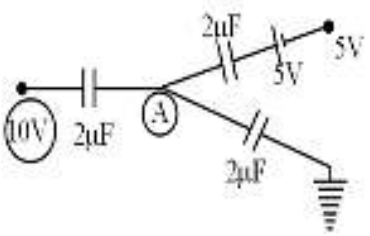
17) A spherical capacitor with outer shell earthed has inner radius R and outer radius 2R. Its capacitance is :-

(1) $4\pi\varepsilon_0 R$

(2) $2\pi\varepsilon_0 R$

(3) $8\pi\varepsilon_0 R$

(4) $6\pi\varepsilon_0 R$



18)
Potential of point A is :-

(1) $\frac{5}{3}V$

(2) 4V

(3) $\frac{20}{3}V$

(4) 2 V

19) A closely wound flat circular coil of 25 turns of wire has diameter of 10 cm and carries a current of 4 ampere. Determine the flux density at the centre of a coil :-

(1) 1.679×10^{-5} tesla

(2) 2.028×10^{-4} tesla

(3) 1.257×10^{-3} tesla

(4) 1.512×10^{-6} tesla

20) If a particle of charge 10^{-12} coulomb, moving along the $\hat{x}$-direction with a velocity 10^5m/s, experiences a force of 10^{-10} newton in $\hat{y}$- direction due to magnetic field, then the minimum magnetic field is :-

(1) 6.25×10^3 tesla in negative $\hat{z}$-direction

(2) 10^{-15} tesla in $\hat{z}$-direction

(3) 6.25×10^{-3} tesla in $\hat{z}$-direction

(4) 10^{-3} tesla in negative $\hat{z}$-direction

21) A current i is flowing in a straight conductor of length L. The magnetic induction at a point distant $\frac{L}{4}$ (Perpendicular to length) from its centre will be :-

(1) $\frac{4\mu_0 i}{\sqrt{5}\pi L}$

(2) $\frac{\mu_0 i}{2\pi L}$

(3) $\frac{\mu_0 i}{\sqrt{2}\, L}$

(4) Zero

22) A long magnet is cut in two parts in such a way that the ratio of their lengths is 2 : 1. The pole strengths of both the section will be :-

(1) In the ratio of 1 : 1

(2) In the ratio of 2 : 1

(3) In the ratio of 1 : 2

(4) In the ratio of 4 : 1

23) For protecting a sensitive equipment from the external magnetic field, it should be :-

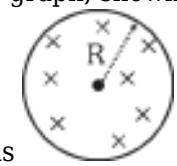
(1) Placed inside an aluminium box

(2) Placed inside an iron box

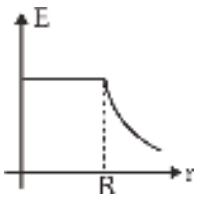
(3) Wrapped with insulation around it when passing current through it

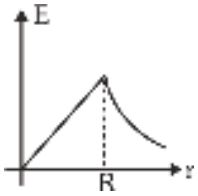
(4) Surrounded with fine copper sheet

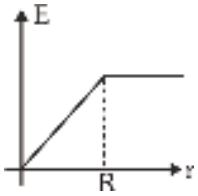
24) A cylindrical space of radius R is filled with a uniform magnetic induction B parallel to the axis of the cylinder. If $\frac{dB}{dt}$ = constant, the graph, showing the variation of induced electric field with distance r from the axis of cylinder, is

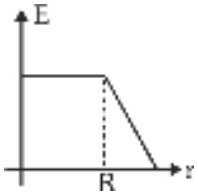


(1)



(2) 

(3) 

(4) 

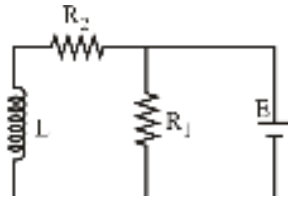
25) A closed coil of copper, whose area is 1 m × 1 m, is free to rotate about an axis. The plane of coil is placed perpendicular to a magnetic field of 0.10 Wb/m^2. It is rotated through 180° in 0.01 s. The average induced emf and average induced current in the coil will, respectively, be (The resistance of the coil is 2.0 Ω)

(1) 20 V, 1 A

(2) 10 V, 10 A

(3) 20 V, 10 A

(4) 5 V, 5 A

26) Find out time constant of the given LR-circuit :- 

(1) $\frac{L}{R_1}$

(2) $\frac{L}{R_2}$

(3) $\frac{L}{R_1 + R_2}$

(4) $\frac{L(R_1 + R_2)}{R_1 R_2}$

27) In an AC circuit, V and I are given by :
V=150sin (150t) V and
I=150 sin$\left(150t + \frac{\pi}{3}\right)$ A.
The power dissipated in the circuit is

(1) Zero

(2) 5625W

(3) 150W

(4) 106W

28) The value of L, C and R for a series circuit are 1H, 9F and 3Ω. What is the quality factor for the circuit at resonance?

(1) 1

(2) 9

(3) $\frac{1}{9}$

(4) $\frac{1}{3}$

29) In series RLC, resonance frequency is 12 KHz. If R = 5Ω and X_L at resonance is 300Ω, The half power frequencies will be :–

(1) 11900 Hz, 12100 Hz

(2) 9000 Hz, 10000 Hz

(3) 200 Hz, 210 Hz

(4) None of the above

30) A single slit of width 0.20 mm is illuminated with light of wavelength 500nm.The observing screen is placed 80 cm from the slit.The width of the central bright fringe will be

(1) 1mm

(2) 2mm

(3) 4mm

(4) 5mm

31) Incident unpolarised light has amplitude A and angle between axis of analyser and polariser is 60°. What will be the amplitude of light coming out finally from polariser?

(1) $\frac{A}{\sqrt{2}}$

(2) $\frac{A}{2\sqrt{2}}$

(3) $A\sqrt{2}$

(4) $\frac{\sqrt{3}}{2}A$

32) An EM wave propagating in +x-direction has a wavelength of 8 mm. The electric field vibrating in +y-direction has maximum magnitude of 60 Vm^{-1}. Choose the correct equations for electric and magnetic fields if the EM wave is propagating in vacuum :

(1) $\begin{matrix} E_y = 60\sin\left[\frac{\pi}{4}\times 10^3\left(x-3\times 10^8 t\right)\right]\hat{j}Vm^{-1} \\ B_z = 2\sin\left[\frac{\pi}{4}\times 10^3\left(x-3\times 10^8 t\right)\right]\hat{k}T \end{matrix}$,

(2) $\begin{matrix} E_y = 60\sin\left[\frac{\pi}{4}\times 10^3\left(x-3\times 10^8 t\right)\right]\hat{j}Vm^{-1} \\ B_z = 2\times 10^{-7}\sin\left[\frac{\pi}{4}\times 10^3\left(x-3\times 10^8 t\right)\right]\hat{k}T \end{matrix}$,

(3) $\begin{matrix} E_y = 2\times 10^{-7}\sin\left[\frac{\pi}{4}\times 10^3\left(x-3\times 10^8 t\right)\right]\hat{j}Vm^{-1} \\ B_z = 60\sin\left[\frac{\pi}{4}\times 10^3\left(x-3\times 10^8 t\right)\right]\hat{k}T \end{matrix}$,

(4) $\begin{matrix} E_y = 2\times 10^{-7}\sin\left[\frac{\pi}{4}\times 10^4\left(x-4\times 10^8 t\right)\right]\hat{j}Vm^{-1} \\ B_z = 60\sin\left[\frac{\pi}{4}\times 10^4\left(x-4\times 10^8 t\right)\right]\hat{k}T \end{matrix}$,

33) A point source of power 15W is placed at a certain point in the space. The amplitude of magnetic field at a distance of 2 meter from source is

(1) 5×10^{-8} T

(2) 6×10^{-8} T

(3) 10×10^{-8} T

(4) 1×10^{-8} T

34) If a thin prism of glass is dipped in water, then minimum deviation of light produced by prism in comparision to air, will become :-

$\left({}_a\mu_g = \frac{3}{2} \text{ and } {}_a\mu_w = \frac{4}{3}\right)$

(1) $\frac{1}{2}$ times

(2) $\frac{1}{4}$ times

(3) 2 times

(4) $\frac{1}{5}$ times

35) The focal length of a convex mirror is 30 cm. Its radius of curvature will be

(1) 30 cm

(2) 15 cm

(3) 60 cm

(4) 120 cm

36) The velocity of light in a medium is half its velocity in air. Find critical angle for this medium-air combination.

(1) 30°

(2) 60°

(3) 45°

(4) 15°

37) Read following statements carefully and select correct answer –
(a) Kinetic energies of emitted photo electrons are not equal even when monochromatic light is used.
(b) Electrons are in different energy levels in a metal.
(c) Work function is related with threshold frequency.
(d) Time interval between incident of photon and emission of electron is 10^{-9} s or less.

(1) Only a, b

(2) a, b, c

(3) b, c, d

(4) All of the above

38) Light of wavelength 500 nm incident on photocathode in photoelectric effect experiment whose work function is 1.2 eV. If anode is kept at 4 volt, find kinetic energy of fastest and slowest electrons reaching anode.

(1) 6.28 eV and 3 eV

(2) 5.28 eV and 4 eV

(3) 3.52 eV and 4 eV

(4) 1.42 eV and 0.5 eV

39) **Assertion (A) :** A single experiment related to nature of light can explain both wave and quantum nature of light.
Reason (R) : In single experiment, only one nature of light can be described.

(1) Both (A) & (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) & (R) are correct but (R) is not correct explanation of (A).

(3) Both (A) & (R) are incorrect.

(4) (A) is incorrect while (R) is correct.

40) A planet P is revolving around sun in an elliptical orbit. Let its speeds at perihelion and apehelion are v_P and v_A and its de-Broglie wavelengths at these positions are λ_P and λ_A respectively, then

(1) $\lambda_P > \lambda_A$

(2) $\lambda_P < \lambda_A$

(3) $\lambda_P = \lambda_A$

(4) Data insufficient

41) If m is the mass of electron, v its velocity, r is the radius of circular orbit around a nucleus with charge Ze; then from Bohr's second postulate, the radius of the quantised orbit in CGS system is given by -

(1) $\frac{4\pi^2 mZe^2}{n^2h^2}$

(2) $\frac{n^2h^2Ze^2}{4\pi^2m}$

(3) $\frac{4\pi^2m}{n^2h^2Ze^2}$

(4) $\frac{n^2h^2}{4\pi^2me^2Z}$

42) Read the following statements carefully and select the correct option.
Assertion (A) :- According to classical theory the proposed path of an electron in Rutherford atom model will be parabolic.
Reason (R) :- According to electromagnetic theory an accelerated charged particle continuously emits radiation.

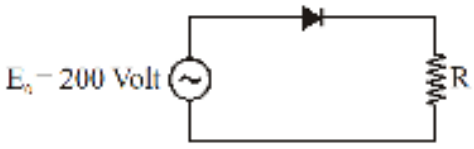
(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true and reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

43) A sinusoidal voltage of peak value 200 volt is connected to a diode and resistor R in the circuit shown so that half wave rectification occurs. If the forward resistance of the diode is negligible compared to R, then rms voltage (in volt) across R is approximately -



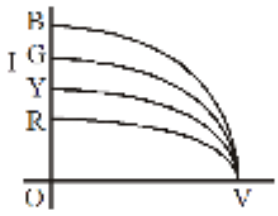
(1) 200

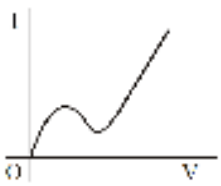
(2) 100

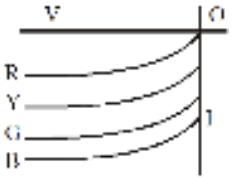
(3) $\frac{200}{\sqrt{2}}$

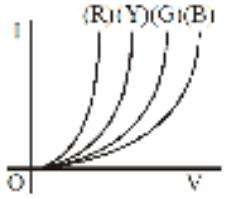
(4) 280

44) The I-V characteristic of a LED is

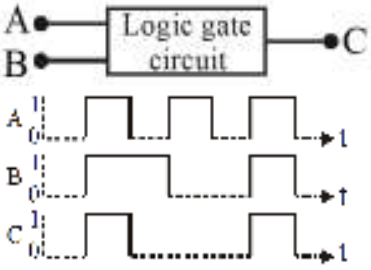
(1)


(2)


(3) 

(4) 

45) The following figure shows a logic gate circuit with two inputs A and B and output C. The voltage waveforms of A, B and C are as shown in second figure below. The logic gate is :



(1) OR gate

(2) NAND gate

(3) AND gate

(4) NOR gate

CHEMISTRY

1) Match the column :-

	Complex name		**Property**
(P)	Ziese's salt	(I)	Obeys sidgwick EAN rule
(Q)	Sodium Nitroprusside	(II)	Heteroleptic complex
(R)	Prussian Blue	(III)	Oxidation state of central metal ion is +2
(S)	Brown ring test complex	(IV)	Example of π bonded organometallic compound

(1) P → II, III, IV, Q → I, III, R → I, III, S → II

(2) P → II, III, Q → I, III, R → I, III, S → II, III

(3) P → I, IV, Q → II, R → II, III, IV, S → I, III

(4) P → II, III, IV, Q → I, II, III, R → I, III, S → II

2) How many of the following species are diamagnetic in nature.
$[Ni(PPh_3)_4]$, $[Cu(NH_3)_4]SO_4$, $[Fe(H_2O)_5NO]SO_4$, $Cr(gly)_3$, $[Co(en)_3]Cl_3$, $[Fe(CO)_5]$,

$[Ni(NH_3)_6]Cl_2, [Cr(NH_3)_6]Cl_2$

(1) 3

(2) 5

(3) 8

(4) 4

3) **Assertion :** Basicity of phosphonic acid and pyrophosphorous acid are equal.
Reason: Both the compound having similar number of OH group.

(1) Both Assertion & Reason are correct and Reason is correct explanation of Assertion.

(2) Both Assertion & Reason are correct but Reason is not correct explanation of Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect but Reason is correct.

4) Which of the following gives yellow, white, Blue and Red colour respectively during confirmation test of basic radical?

(I)  (II) 

(III)  (IV) 

(1) IV, I, II, III

(2) I, II, III, IV

(3) II, III, I, IV

(4) IV, III, I, II

5) Choose incorrect option:-

(1) $SO_2 > SeO_2 > TeO_2$ Reducing property

(2) $NF_3 > NCl_3 > NBr_3$ Stability

(3) $GeO_2 > SnO_2 > PbO_2$ Oxidising Power

(4) $H_3PO_4 > H_3PO_3 > H_3PO_2$ pKa value

6) Which of the following is an correct order of λ_{absorb} of the following complexes?
I. $[Co(EDTA]^-$ II. $[Co\ (dien)_2]^{+3}$ III. $[Co(en)_3]^{+3}$

(1) I > II > III

(2) III > II > I

(3) II > I > III

(4) I > III > II

7) Choose correct statement about the property of inert gasses.
(I) They are colourless, odourless and tasteless
(II) They are sparingly soluble in water.
(III) Helium has the lowest boiling point among of any known substance.
(IV) Helium show's maximum diffusing property.
(V) Their atmospheric abundance in dry air is ~1% by volume of which xenon is major constituent.

(1) I, II

(2) I, II, III, IV

(3) II, III, V

(4) I, III, V

8) Which of the following is not a green colour compound?

(1) $FeSO_4 (NH_4)_2 SO_4, 6H_2O$

(2) $K_2SO_4 Al_2(SO_4)_3. 24H_2O$

(3) $NiCl_2$ (Aqueous)

(4) $Cr(OH)_3$

9) Which of the following pair of species show's geometrical as well as optical isomerism?

(1) $[Co(en)_3]Cl_3$ & $[Co(en)_2Cl_2]^+$

(2) $[Zn (NH_3)_2Cl_2]$ & $[Pt (NH_3)_2Cl_2Br_2]$

(3) $[Cr(gly)_3]$ & $[Fe(gly) (NH_3)_2Cl_2]^-$

(4) $[Cr(NH_3)_3Cl_3]$ & $[Pt(NH_3) (Py) (Cl) (Br) (I) (OH)]$

10) Mixture of salt gives following reactions



It confirm the presence of

(1) Pb^{+2}

(2) Hg_2^{+2}

(3) Ag^+

(4) None of these

11) The nature of Nitric oxide and Nitrogen sesquioxide are respectively?

(1) Acidic & Acidic

(2) Amphoteric & Acidic

(3) Neutral & Acidic

(4) Neutral & Neutral

12) Which of the following contain peroxy linkage?

(1) $H_2S_2O_8$

(2) $H_2S_2O_5$

(3) $H_2S_2O_7$

(4) $H_2S_4O_6$

13) In which of the following conversion central atom doesn't change its oxidation state?

(1) $Na_2[Fe(CN)_5NO] \xrightarrow{+Na_2S}$ 'x'

(2) $CuSO_4 \xrightarrow{+KI}$ 'y'

(3) $FeCl_3 \xrightarrow{+KI}$ 'z'

(4) $FeSO_4(aq.) \xrightarrow{+NO}$ 'w'

14) The Lake test confirmed the presence of

(1) Al^{+3}

(2) Fe^{+3}

(3) Ba^{+2}

(4) Cu^{+2}

15) Which of the following statements are incorrect?
(a) K_2MnO_4 & $KMnO_4$ coloured due to d–d transition
(b) In a basic medium V_2O_4 forms VO_2^+
(c) Generally transition element forms alloy due to their comparable size.
(d) In a 3d series Fe, Mn, Co can form's mixed oxide.

(1) a, b, c

(2) a, b, c, d

(3) c, d

(4) a, b

16) For two first order reactions having same concentration of A and B at t = 0, use the data given below to calculate the temperature at which both reaction occurs with same rate :-
Reaction-I $A \rightarrow Z$, $K_1 = 10^{16}e^{(-3000/T)}$
Reaction-II $B \rightarrow Y$, $K_2 = 10^{15}e^{(-2000/T)}$

(1) 434.22 K

(2) 1446 K

(3) 727.1 K

(4) 981.22 K

17) Which of the following is incorrect for an ideal solution ?

(1) $\Delta H_{mix} = 0$

(2) $\Delta V_{mix} = 0$

(3) $\Delta S_{mix} = 0$

(4) $P_A = P^o_A X_A$

18) What will be the osmotic pressure of 0.2 M $K_4[Fe(CN)_6]$ at 27°C ?

(1) 24.6 atm

(2) 49.2 atm

(3) 4.92 atm

(4) 2 atm

19) The equivalent conductivity at infinite dilution (Λ^∞) for sodium chloride, HCl and sodium acetate in water at 298 K are 128.0, 425.0 and 91.0 $Scm^2\ eq^{-1}$ respectively. A solution of acetic acid shows an equivalent conductivity of 38.8 $Scm^2\ eq^{-1}$. What is the percent dissociation of acetic acid.

(1) 5.0

(2) 10.0

(3) 20.0

(4) 30.0

20) Mechanism of a hypothetical reaction :
$A_2 + B_2 \rightarrow 2AB$ is given below
(i) $A_2 \rightleftharpoons A + A$ (Fast)
(ii) $A + B_2 \rightarrow AB + B$ (Slow)
(iii) $A + B \rightarrow AB$ (Fast)
The overall order of the reaction will be :

(1) 2

(2) 0

(3) 1.5

(4) 1

21) A first order reaction has a rate constant of $4.62 \times 10^{-3}\ s^{-1}$. The time required for 0.4 g of this reactant to reduce to 0.1 g will be :

(1) 230.3 s

(2) 200 s

(3) 300 s

(4) 23.03 s

22) **Assertion (A) :** In a reversible endothermic reaction, activation energy of the forward reaction is higher than that of the backward reaction.
Reason (R) : In a reversible endothermic reaction threshold energy of the forward reaction is more than that of the backward reaction.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**.

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is incorrect but **(R)** is correct.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.

23) The following data is given for reaction between A and B :-

S. No.	**[A] mol. L^{-1}**	**[B] mol. L^{-1}**	**Rate mol. L^{-1} s^{-1}**
I	1×10^{-2}	2×10^{-2}	2×10^{-6}
II	2×10^{-2}	2×10^{-2}	4×10^{-6}
III	2×10^{-2}	4×10^{-2}	8×10^{-6}

Which of the following are correct conclusions from above data ?
(a) Rate constant of the reaction is 10^{-2} s^{-1}
(b) Rate law of the reaction is, r = k[A] [B]
(c) Rate of reaction increases nine times when the concentration of both the reactant are tripled.

(1) a, b and c

(2) a and b

(3) b and c

(4) c only

24) If degree of dissociation of CH_3COOH in its aqueous solution is 0.1 and Λ_m^∞ for NaCl, HCl and CH_3COONa are 125, 425 and 90 S cm^2 mol^{-1} respectively, then conductivity of 0.001 M CH_3COOH aqueous solution is -

(1) 3.9×10^{-5} S cm^{-1}

(2) 1.5×10^{-5} S cm^{-1}

(3) 8.2×10^{-5} S cm^{-1}

(4) 6.7×10^{-5} S cm^{-1}

25) The plot of lnk vs $\frac{1}{T}$ for the following reaction
$2NH_{3(g)} \rightarrow N_{2(g)} + 3H_{2(g)}$ gives a straight line with the slope of line equal to -3.0×10^{-5} K. Activation energy for the reaction in J mol^{-1} is -
(R = 8.3 J K^{-1} mol^{-1})

(1) 1.7×10^{-4}

(2) 8.6×10^{-4}

(3) 4.1×10^{-4}

(4) 2.5×10^{-4}

26) At a certain temperature, 45 g of glucose present in 1 litre of its solution, has an osmotic pressure of 5 atm. If the osmotic pressure of the other solution is 2 atm at the same temperature, what would be its concentration ?

(1) 0.2 mol/litre

(2) 0.1 mol/litre

(3) 0.28 mol/litre

(4) 0.5 mol/litre

27) 5g of an unknown solute is dissolved in 295 g solvent. If molarity and density of solution are 0.05 M and 1.5 g/cc respectively. The molecular weight of unknown solute is :

(1) 375

(2) 425

(3) 400

(4) 500

28) The emf of cell
Pt | $H_2(P_1)$ | $H^+(aq)(1M)$ || $H^+(aq)(1M)$ | $H_2(P_2)$ | Pt,
is :

(1) $\frac{RT}{F}\ln\frac{P_1}{P_2}$

(2) $\frac{RT}{2F}\ln\frac{P_1}{P_2}$

(3) $\frac{RT}{F}\ln\frac{P_2}{P_1}$

(4) $\frac{RT}{2F}\ln\frac{P_2}{P_1}$

29) **Assertion :** In galvanic cell, chemical energy is converted into electrical work by spontaneous redox reaction.
Reason : In galvanic cell, electrons flow from cathode to anode in external circuit.

(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(2) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

30) Correct matching of column I with column II is :

	Column-I		**Column-II**
(i)	Lead storage battery	(a)	gives steady potential
(ii)	Mercury cell	(b)	Pb is anode, PbO_2 is cathode
(iii)	Fuel cell	(c)	Cell reaction : $2H_2(g) + O_2(g) \rightarrow 2H_2O(\square)$

(1) (i) - (a), (ii) - (c), (iii) - (b)

(2) (i) - (c), (ii) - (a), (iii) - (b)

(3) (i) - (b), (ii) - (a), (iii) - (c)

(4) (i) - (c), (ii) - (b), (iii) - (a)

31) Which can give haloform with X_2/OH^- ?

(1) CH_3OH

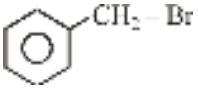
(2) $CH_3CH_2CH{=}O$

(3) CH_3CHO

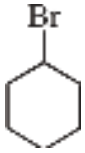
(4) $CH_3CH_2CH_2OH$

32)

Compare the rate of S_N1 reaction

(I) 

(II) 

(III) 

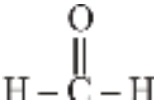
(1) I > II > III

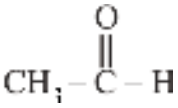
(2) II > III > I

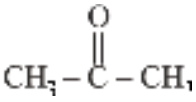
(3) II > I > III

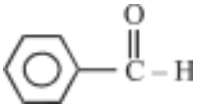
(4) III > I > II

33) Which of the following compound when reacts with CH_3MgBr form 3° alcohol?

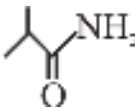
(1) 

(2) 

(3) 

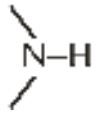
(4) 

34) Which one among the following is expected to form a secondary alcohol on treatment with HNO_2 ?

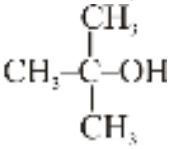
(1) 

(2) 

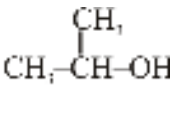
(3) 

(4) 

35) Arrange the following in order of their reactivity when reacting with $HCl/ZnCl_2$:-



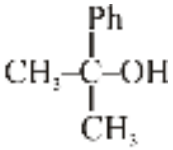
I



II



III



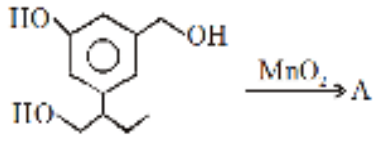
IV

(1) IV > I > II > III

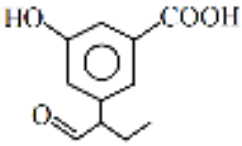
(2) I > II > III > IV

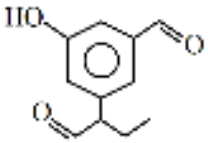
(3) IV > III > II > I

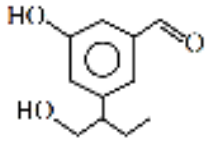
(4) III > II > I > IV

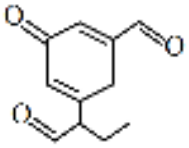
36) 

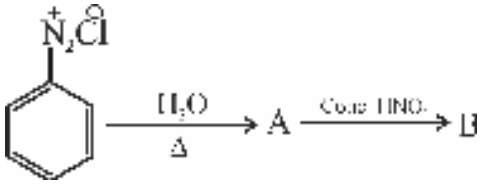
The product "A" is :-

(1) 

(2) 

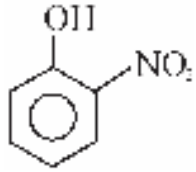
(3) 

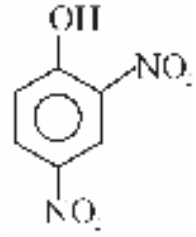
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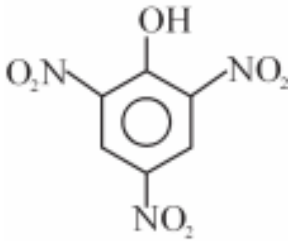
37) 

Product B is :

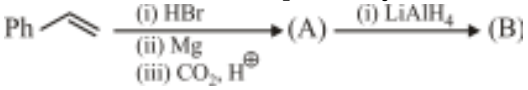
(1) 

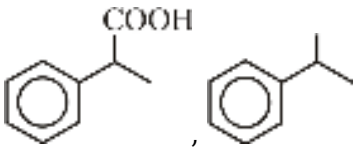
(2) 

(3) 

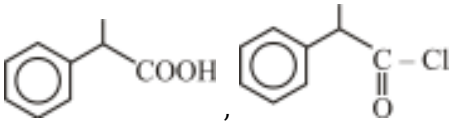
(4) 

38) Product A & B respectively :-

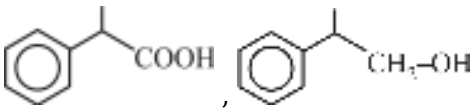


(1) 

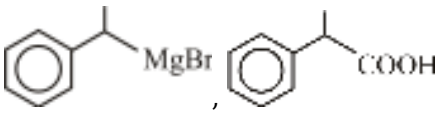
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(2) 

,

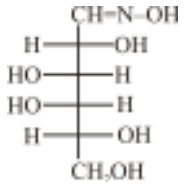
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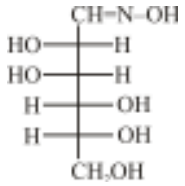
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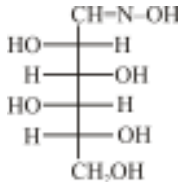
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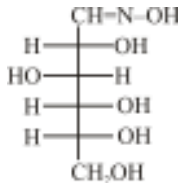
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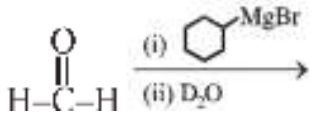
39) D(+) Glucose react hydroxylamine (NH_2OH) to give oxime. Structure of oxime is :

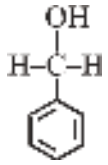
(1) 

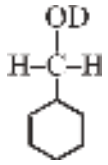
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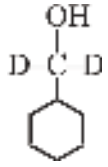
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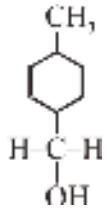
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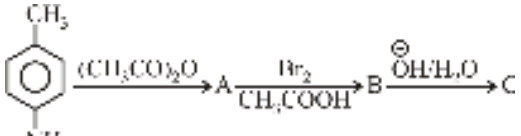
40)  [A] ? A is :

(1) 

(2) 

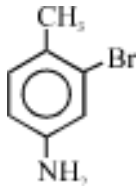
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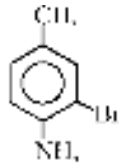
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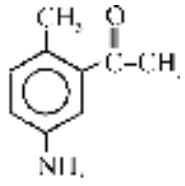
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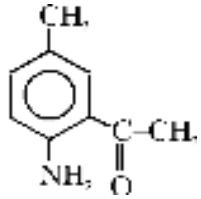
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Product C will be :-

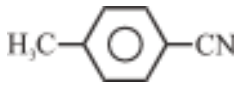
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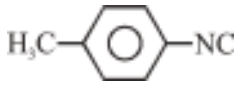
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(3) 

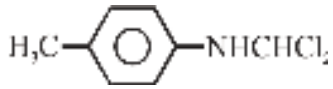
(4) 

42) The reaction of chloroform with alcoholic KOH and p-toluidine forms :

(1) 

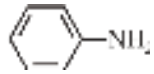
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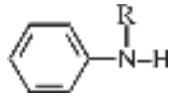
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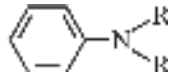
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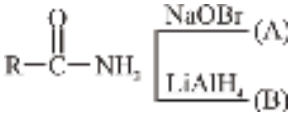
43) Which of the following amines from N–nitroso derivative when treated with $NaNO_2$ and HCl ?

(1) CH_3NH_2

(2) 

(3) 

(4) 

44)  relation between A and B is :-

(1) Position Isomers

(2) Homologues

(3) Functional Isomers

(4) Identical

45) Match list I with II and choose the correct answer from the codes given below:-

List-I (Compound)		**List-II (Aplication)**	
(A)	Aniline	(a)	Used in Making azo dyes
(B)	Nitrobenzene	(b)	Sulpha Drug
(C)	Sulphanilamide	(c)	Solvent in the friedal crafts reactions
(D)	Trinitrotoluene	(d)	Used as explosive

Code is:-

	A	**B**	**C**	**D**
(1)	a	c	b	d
(2)	a	b	c	d
(3)	c	d	a	b
(4)	d	c	b	a

(1) 1

(2) 2

(3) 3

(4) 4

BIOLOGY

1) What is the pattern of inheritance for polygenic trait?

(1) Mendelian inheritance pattern

(2) Non-mendelian inheritance pattern

(3) Autosomal dominant pattern

(4) X-linked recessive inheritance pattern

2) Broad palm with characteristic palm crease is visible in a person suffering from :-

(1) Down's syndrome

(2) Turner's syndrome

(3) Klinefelter's syndrome

(4) Thalassemia

3) Match the Column-I with Column-II :-

	Column-I		**Column-II**
(A)	A single trait controlled by three or more than three allele	(i)	Pleiotropy
(B)	A single trait controlled by three or more than three genes	(ii)	Multiple allele
(C)	A single gene exhibits multiple phenotypic expression	(iii)	Polygenic inheritance

(1) A - ii, B - iii, C - i

(2) A - iii, B - ii, C - i

(3) A - i, B - ii, C - iii

(4) A - ii, B - i, C - iii

4) Who noted that behaviour of chromosomes was parallel to behaviour of genes and used chromosomal movement to explain Mendel's laws ?

(1) Bateson

(2) Alfred Sturtevant

(3) T.H. Morgan

(4) Sutton and Boveri

5)

Which one of the following can be explained on the basis of Mendel’s Law of Dominance?
A. Out of one pair of factors one is dominant and the other is recessive.
B. Alleles do not show any blending and both the characters appear as such in F_2 generation.
C. Factors occur in pairs in normal diploid plants.
D. The discrete unit controlling a particular character is called factor.
E. The expression of only one of the parental characters is found in F_1 generation of a monohybrid cross.
Choose the correct answer from the options given below :

(1) A, B and C only

(2) A, C, D and E only

(3) B, C and D only

(4) A, B, C, D and E

6) Linkage was described by Morgan in *Drosophila* :
(a) He hybridised yellow-bodied, white-eyed females to brown-bodied, red-eyed males and then

intercrossed their F_1 progeny.
(b) The ratio of F_2 generation is same as that of 9: 3: 3: 1 in this cross.
(c) The parental combinations are observed more than recombinants.
(d) According to Morgan linkage was observed due to chemical association of genes.
(e) All the genes present on same chromosomes have same efficiency to link.
Choose the correct answer from the options given below :

(1) (b), (d), (e)

(2) (a), (c)

(3) (a), (c), (e)

(4) (a), (d)

7) Select the incorrect statement(s) from the options given below with respect to dihybrid cross.
I. Tightly linked genes on the same chromosome show higher recombinations.
II. Genes far apart on the same chromosomes show very few recombinations.
III. Genes loosely linked on the same chromosome show 100% recombinations.

(1) I and II

(2) III and II

(3) I and III

(4) All of these

8)

Match List I with List II :

	List I		**List II**
A.	Down's syndrome	I.	11th chromosome
B.	α-Thalassemia	II.	'X' chromosome
C.	β-Thalassemia	III.	21st chromosome
D.	Klinefelter's syndrome	IV.	16th chromosome

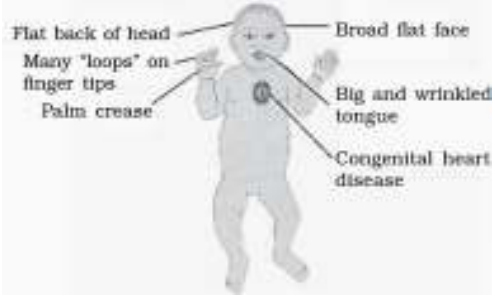
Choose the correct answer from the options given below :
(1) A-I, B-II, C-III, D-IV

(2) A-II, B-III, C-IV, D-I

(3) A-III, B-IV, C-I, D-II

(4) A-IV, B-I, C-II, D-III

9) Given figure depicts.



(1) Syndrome due to aneuploidy of autosome

(2) Trisomy of 21st chromosome

(3) Individual have 47 chromosomes

(4) All of these

10) Which of the following statement(s) is/are true with respect to sickle-cell anaemia ?

(1) The mutant haemoglobin of sickle-cell anaemic individual undergo polymerisation under low oxygen tension causing sickleing of RBCs.

(2) Sickle-cell anaemia occur due to the single base substitution (GAG → GUG) at the sixth codon of β-globin gene.

(3) Individuals heterozygous for sicke-cell anaemia (HbSHbA) are resistant towards malaria.

(4) All of the above

11) A DNA sequence is given: 5'-ATG AAG TCG TAA-3' After mutation, one nucleotide "A" is added after the first codon (ATG). What will be the result of this mutation?

(1) Only one amino acid will change

(2) No effect due to degeneracy of code

(3) Frameshift mutation leading to completely altered protein sequence

(4) Formation of identical protein

12) Which of following DNA strand required many RNA primer at the time of DNA replication :-

(1) Leading strand

(2) Lagging strand

(3) Both

(4) Continuous-strand

13) DNA strand is directly involved in the synthesis of all of the following except :-

(1) t-RNA molecule

(2) Another DNA strand

(3) m-RNA molecule

(4) Protein

14) The packaging of chromatin in eukaryotes at higher level requires set of proteins that are collectively referred to as :-

(1) Histones

(2) Polyamines

(3) Non-Histone Chromosomal protein

(4) Globins

15) The sequence of events mentioned below are symbolized by alphabets. Choose the correct answer where the alphabets are matched with the processes:

RNA $\xrightarrow{(A)}$ DNA $\xrightarrow{(B)}$ DNA $\xrightarrow{(C)}$ m-RNA $\xrightarrow{(D)}$ Polypeptide

(1) A-Replication, B-Transformation, C-Transcription, D-Translation

(2) A-Reverse transcription, B-Translation, C-Transcription, D-Replication

(3) A-Replication, B-Transcription, C-Translation, D-Tranduction

(4) A-Reverse transcription, B-Replication, C-Transcription, D-Translation

16) Which statement about mini-satellites in DNA fingerprinting is **incorrect**?

(1) Mini-satellites consist of small DNA sequences arranged in tandem repeats.

(2) The copy number of mini-satellites varies from chromosome to chromosome within an individual.

(3) The size of mini-satellites is relatively consistent, typically around 10 kb.

(4) Mini-satellites show a high degree of polymorphism, resulting in varying band patterns on autoradiograms.

17) Main steps in DNA fingerprinting are given below. Arrange these steps in a correct sequence.
(A) Autoradiography
(B) Hybridisation
(C) Blotting
(D) Gel electrophoresis
(E) Isolation of DNA
Choose the correct answer from the option given below :-

(1) E,C,B,A,D

(2) E,B,A,C,D

(3) E,D,B,C,A

(4) E,D,C,B,A

18) Read the following statements.
I. A purine is heterocyclic, 9-membered double ring structure with nitrogen at 1st, 3rd, 7th and 9th positions.
II. A pyrimidine is heterocyclic, 6-membered single ring structure with nitrogen at 1st and 3rd positions.
III. Purine nucleosides have 1′-9-glycosidic linkage whereas pyrimidine nucleosides have 1′-1-glycosidic linkage.
IV. Two nucleosides are linked by 3′-5′ phosphodiester linkage to form a dinucleotide.
V. Ribose sugar can be represented as $C_5H_{10}O_4$ whereas deoxyribose sugar can be represented as $C_5H_{10}O_5$.
Which of the above statement(s) is/are correct?

(1) Only I

(2) I, II, III and IV

(3) III, IV and V

(4) IV and V

19) Identify the correct statements about Genetic codons.
A. AUG is initiator codon and codes for Glycine
B. UAA codes for Tyrosine
C. The codons are mostly universal
D. UAG is terminator codon
E. More than one codons code for single amino acid

(1) A, D, E

(2) C, D, E

(3) A, B, C

(4) C, B, D

20) Match the list I with list II and choose the correct answer from the option givens below.

	List-I		List-II
A	Termination	I	Aminoacyl tRNA synthetase
B	Translation	II	Okazaki fragments
C	Transcription	III	GTP dependent release factor
D	Replication	IV	RNA polymerase

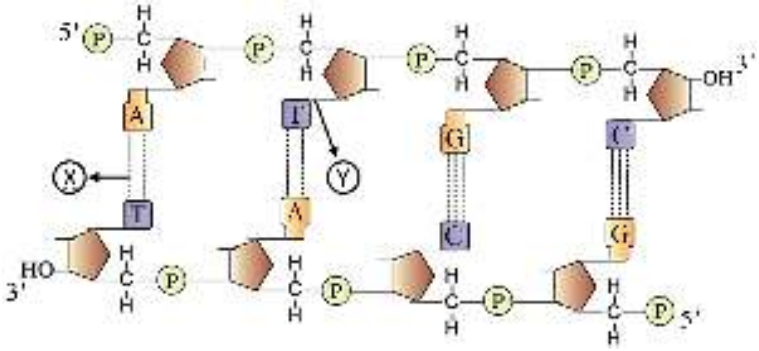
(1) A - II, B - III, C - I, D - IV

(2) A - III, B - I, C - II, D - IV

(3) A - III, B - I, C - IV, D - II

(4) A - II, B - IV, C - I, D - III

21) A polynucleotide chain is shown in the figure. Identify X and Y (types of bond)



	X	Y
1	Glycosidic bond	Hydrogen bond
2	Glycosidic bond	Ionic bond
3	Ionic bond	Covalent bond
4	Hydrogen bond	Glycosidic bond

(1) 1

(2) 2

(3) 3

(4) 4

22) Sigmoid growth curve is represented by :-

(1) dN/dt = rN

(2) dN/dt = rN (K – N/K)

(3) Nt = No + [B + I – D – E]

(4) dN/dt = 1 – N/K

23) "Sexual deceit" is a term used in interspecific interaction some time. This term applies to which of the following.

(1) Honey bee & Orchid

(2) Fig and Wasp

(3) Ophrys and Orchid flower

(4) Bumble bee and Ophrys orchid

24) Match the list-I with list-II and select the correct option given below :

	List-I (Interactions)		**List-II (Species)**
(A)	Mutualism	(i)	+ 0
(B)	Commensalism	(ii)	– 0
(C)	Amensalism	(iii)	+ –
(D)	Parasitism	(iv)	+ +

	A	**B**	**C**	**D**
(1)	ii	iv	iii	i
(2)	iv	i	ii	iii
(3)	iv	ii	i	iii
(4)	ii	iii	iv	i

(1) 1

(2) 2

(3) 3

(4) 4

25) **Assertion (A) :** Predators in nature are too efficient and over exploits its prey.
Reason (R) : Predators in nature are not prudent.

(1) Assertion & Reason both are true and Reason is correct explanation of Assertion.

(2) Assertion & Reason both are false.

(3) Assertion is true but Reason is false

(4) Assertion & Reason both are true and Reason is not the correct explanation of Assertion.

26) Consider the following four statements and select the option which includes all the correct ones:-
(a) Sparrow eating any seed is not less than a predator.
(b) Monarch butterfly is highly distasteful to its predator.
(c) Nearly 25% of all insects are known to be phytophagous.
(d) Thorns are the most common morphological defence mechanism.
Option :-

(1) Statements- (a), (b), (c), (d)

(2) Statements- (a), (b)

(3) Statements- (a), (b), (c)

(4) Statements- (b), (d)

27) More conservative and scientifically sound estimate made by Robert May place global species diversity at about_____.

(1) 70 million

(2) 7 billion

(3) 7 lac

(4) 7 million

28) Find out the correct match from the following table :

	Column-I	**Column-II**	**Column-III**
(i)	In situ conservation	Hot spot	Endemic species
(ii)	Ex situ conservation	Zoological park	Critically endangered
(iii)	Narrowly utilitarian	Pollination	Direct economic benefit

(1) i & ii

(2) i only

(3) ii only

(4) ii & iii

29) **Statement-I :** National park, Wild life sanctuaries and Biosphere reserves are the example of Ex-situ conservation.
Statement-II : Exploring molecular, genetic and species level diversity for the products of economic importance known as bioprospecting.

(1) Statement I and II both are correct

(2) Statement I and II both are incorrect

(3) Only Statement I is correct

(4) Only Statement II is correct

30) Identify correct statements w.r.t biodiversity hotspots.
a. Initially 25 biodiversity hot spots were identified in the world.
b. Now number of biodiversity hotspots are 32.
c. India have 3 hot spots i.e. Eastern Ghat, Indo Burma and the Himalayas.
d. Hot spots are example of In situ conservation.
e. All the biodiversity hotspots put together cover less than 2% of the earth's land area.
Choose option which includes incorrect statements only.

(1) a, d and e

(2) a, b, d and e

(3) a and e

(4) b and c

31) Which of the following is the unit of productivity of an ecosystem ?

(1) gm^{-2}

(2) KCal m^{-2}

(3) KCal m^{-3}

(4) (KCal m^{-2})yr^{-1}

32) The annual net primary productivity of the whole biosphere is :-

(1) 10 billion tons

(2) 170 billion tons

(3) 55 billion tons

(4) 1 billion tons

33) **Assertion :** The productivity of terrestrial ecosystem is more than aquatic ecosystem.
Reason : The number of producers are more in terrestrial ecosystem compare to aquatic ecosystem.

(1) Both Assertion and reason are correct and reason is not correct explanation of assertion.

(2) Both Assertion and reason are correct and reason is correct explanation of assertion.

(3) Assertion is correct and reason is incorrect.

(4) Assertion is incorrect and reason is correct.

34) Given below are two statements :
Statement-I :- Detritivores break down detritus into smaller particles this process is called fragmentation.
Statement-II :- Decomposition is faster if the detritus is rich in lignin and chitin.
Choose the correct answer from the option the option the options given below :

(1) Both statement I and statement II are incorrect

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are correct

35)

Tropical regions show greatest level of species richness because -
A. Tropical latitudes have remained relatively undisturbed for millions of years, hence more time was available for species diversification.
B. Tropical environments are more seasonal.
C. More solar energy is available in tropics.
D. Constant environments promote niche specialization.
E. Tropical environments are constant and predictable.
Choose the correct answer from the options given below :

(1) A, C, D and E only

(2) A and B only

(3) A, B and E only

(4) A, B and D only

36) Which statement is incorrect ?

(1) Trophic level represent functional level not a species as such.

(2) A same species may occupy more than one trophic level in same ecosystem at same time.

(3) In grass land ecosystem, all pyramids of number, of energy and biomass are upright.

(4) Energy at higher trophic level is more as compared to lower levels as they are physically strong.

37) Which of the following statement is correct?

(1) Nucellar adventive embryony is found in citrus and mango

(2) In grass family, the radical enclosed in a hollow foliar structure called coleoptile

(3) Syngamy lead to formation of a primary endosperm cell

(4) All aquatic plants use water for pollination

38) Identify the incorrect statement :-

(1) Pollination by wind is the most common amongst abiotic pollinations.

(2) Wind Pollination also requires that the pollen grains are light and Non-sticky.

(3) Corn cob-the tassels you see are nothing but the stigma & style

(4) Insect - pollination is quite common in grasses.

39) Choose the correct match from following table :

	Column-I	**Column-II**
(A)	Coleorhiza	Shoot apex
(B)	Scutellum	Cotyledon in grass family

(C)	Primary Endosperm nucleus	Diploid
(D)	Pericarp	Outer layer of anther

(1) B & C

(2) only C

(3) C, B & A

(4) only B



40) Identify the correct description about the given figure :

(1) Wind pollinated plant inflorescence showing flowers with well exposed stamens.

(2) Water pollinated flowers showing stamens with mucilaginous covering.

(3) Cleistogamous flowers showing autogamy

(4) Compact inflorescence showing complete autogamy.

41) How many of the following statements are true ?
(i) Residual, persistant nucellus is the perisperm.
(ii) 2000 years old viable seeds of date palm excavated from arctic tundra.
(iii) Central cell after triple fusion becomes primary endosperm cell.
(iv) Emasculation is not needed if the female parent produces unisexual flowers.
Options -

(1) Four

(2) Three

(3) Two

(4) One

42) Which of the following structure is not present in germ pore?

(1) Sporopollenin

(2) Pectocellulose

(3) Cellulose

(4) Protein

43) Pollen grains remain viable for months in-

(1) Rosaceae and Solanaceae

(2) Poaceae and Leguminosae
(3) Brassicaceae and Poaceae
(4) Brassicaceae and Solanaceae

44) The central cell after triple fusion becomes the

(1) Megaspore
(2) Zygote
(3) Primary endosperm cell
(4) Synergid

45) All of the following are advantage of seeds in plants except-

(1) Provide mechanism of pollination
(2) Protection to embryo
(3) Means of genetic variation
(4) Nourishment to embryo

46) Agarose extracted from sea weeds is used in -

(1) Spectrophotometry
(2) Tissue culture
(3) PCR
(4) Gel electrophoresis

47) Which of the following is not a property of good cloning vector DNA :-

(1) Origin of replication (ori)
(2) It contain a selectable marker
(3) Single recognition site for commonly used restriction enzyme
(4) Many recognition sites for commonly used restriction enzyme

48) If recombinant DNA is inserted within the coding sequence of enzyme galactosidase, which of the following will occur in case of non-recombinants:-

(1) Insertional inactivation
(2) Colonies do not produce any colour
(3) Chromogenic substrate gives blue colour
(4) Inactivation of enzyme galactosidase

49) Which of the following is a vector based method to transform plants?

(1) Retroviral
(2) Ti Plasmid
(3) Micro - injection
(4) Biolistic

50) **Assertion :** Each restriction endonuclease functions by inspecting the length of a DNA sequence.
Reason : Each restriction endonuclease recognises a specific nucleotide sequence in the DNA.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

51) Match the columns :

	Column-I		**Column-II**
(a)	PCR	(i)	Amplification of gene
(b)	Bioreactor	(ii)	Large scale culture
(c)	Gene gun	(iii)	Restriction endonuclease
(d)	EcoRI	(iv)	Transfer DNA into plant cell.

(1) a-i, b-iv, c-ii, d-iii
(2) a-i, b-ii, c-iv, d-iii
(3) a-ii, b-i, c-iv, d-iii
(4) a-iii, b-iv, c-i, d-ii

52) **Statement-1 :** The ligation of alien DNA is carried out at a restriction site present in one of the two antibiotic resistance gene.
Statement-2 : Selection of recombinants due to activation of antibiotics is a cumbersome procedure because it requires simultaneous plating on two plates have different antibiotics.

(1) Both Statements 1 and 2 are correct.
(2) Both Statements 1 and 2 are incorrect.
(3) Statement 1 is correct but 2 is incorrect.
(4) Statement 1 is incorrect but 2 is correct.

53) Read the following statements :-
(a) Enzyme linked immunosorbent (ELISA) is based on the principle of antigen antibody interaction.
(b) ADA enzyme is crucial for the immune system to function.
(c) In PCR multiple copies of the gene or DNA of interest is synthesised in vivo using two set of primer.
(d) In polymerase chain reaction two sets of RNA primers is used.
(e) During PCR repeated amplification is achieved by the use of a thermostable DNA polymerase.
Choose the option with correct statements :-

(1) a, b & c
(2) c, d & e
(3) a, b, c, d, e
(4) a, b & e

54) Virus-free plants can be obtained through :-

(1) Meristem culture
(2) Embryo culture
(3) Protoplast culture
(4) Anther culture

55) Read the given following statements :-
(a) Insulin consists of two short polypeptide chains
(b) ELISA is based on the principle of antigen-antibody interaction
(c) cryIAB controls corn borer
(d) ADA enzyme is crucial for the immune system to function
(e) C peptide is present in the mature insulin
Choose the correct answer from the options given below :-

(1) (b), (d), (e) only
(2) (a), (b), (c), (d)
(3) (a), (c), (e) only
(4) (a), (c) only

56) The use of bioresources by multinational companies and other organization without proper authorisation from countries and people concered with compensatory payment it is known as :-

(1) Biopiracy
(2) Biopatent
(3) Biowar
(4) Bioremediation

57) **Assertion (A) :** In 1997, an American Company got patent right on Basmati rice through the US patent and trademark office.
Reason (R) : Basmati Rice is distinct for its unique aroma and flavour and 27 documented varieties of Basmati are grown in India.

(1) Both A and R are correct but R is not the correct explanation of A.
(2) Both A and R are correct but R is correct explanation of A
(3) A is correct but R is not correct
(4) A is incorrect but R is correct

58) Match the column I, II & III.

	Column-I		**Column-II**		**Column-III**
A	Golden rice	P	ADA deficiency	I	Cystic fibrosis
B	Tobacco	Q	Study of disease	II	Vitamin A enriched
C	Gene therapy	R	Enhanced nutritional value	III	RNA interference

D	Transgenic animal	S	*Meloideyne incognita*	IV	Genetically engineered lymphocytes

Choose the correct answer from the options given below.

(1) A - R - II, B - S - III, C - P - IV, D - Q - I

(2) A - Q - I, B - P - III, C - S - II, D - R - IV

(3) A - P - I, B - S - II, C - R - IV, D - Q - III

(4) A - S - IV, B - Q - I, C - R - II, D - P - III

59) Biological process is carried out in which type of water treatment :

(1) Primary

(2) Secondary

(3) Tertiary

(4) None of above

60) Select the group of organisms that are used as biofertilisers in organic farming.

(1) *Clostridium, Glomus* and *Anabaena*

(2) *Trichoderma, Baculovirus* and *B. thuringiensis*

(3) *Nostoc, Azolla* and Tobacco mosaic virus

(4) *Penicillium, Streptococcus* and *Aspergillus*

61) How many of the following bioactive chemicals are produced by certain type of unicellular fungi ?
Citric acid, Butyric acid, Statins, Cyclosporin-A, Streptokinase, Acetic acid, Ethanol

(1) Three

(2) Four

(3) Five

(4) Two

62) Which of the following statement is correct ?

(1) Biogas is a mixture of gases produced by the microbial activity and which may be used as fuel.

(2) The biogas plant consists of a concrete tank (10-15 feet deep) in which bio-wastes are collected and a slurry of dung is fed.

(3) The biogas thus produced is used for cooking and lighting.

(4) All of the above

63) **Assertion :-** Parturition is induced by a complex neuroendocrine mechanism.
Reason :- The signals from fully developed foetus and placenta which induce mild uterine contraction called foetal ejection reflex this trigger release of oxytocin.

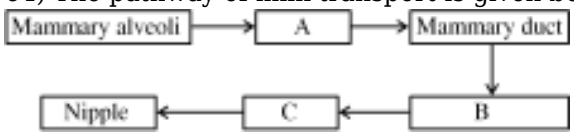
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

64) The pathway of milk transport is given below.



Identify the option in which (a), (b) and (c) are given correctly :-

	A	B	C
(1)	Mammary ampulla	Mammary tubules	Lactiferous duct
(2)	Lactiferous duct	Mammary ampulla	Mammary tubules
(3)	Mammary tubules	Mammary ampulla	Lactiferous duct
(4)	Lactiferous duct	Mammary tubules	Mammary ampulla

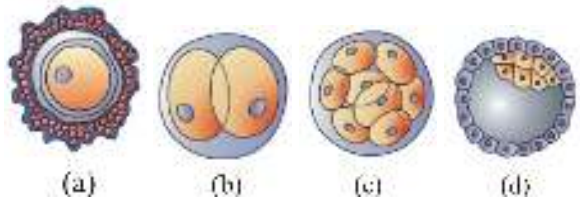
(1) 1

(2) 2

(3) 3

(4) 4

65) The figure below show four stage (a, b, c, d) of human development. Select the option giving correct identification together with site of occurrence ?



		Developmental stage	Site of occurrence
(1)	(a)	Fertilised egg	Isthms part of fallopian tube
(2)	(b)	Zygote	Distal part of fallopian tube
(3)	(c)	Morula	Starting point of fallopian tube
(4)	(d)	Blastocyst	Uterine cavity

(1) 1

(2) 2

(3) 3

(4) 4

66) Identify the correct match from the columns I, II and III :-

Column-I		Column-II		Column-III	
A	Leydig cells	a	Ovary	i	Inhibin
B	Sertoli cells	b	Semini-ferous tubule	ii	Progesterone
C	Theca interna	c	Testis	iii	Androgens
D	Corpus luteum	d	Follicle	iv	Estrogen

(1) A → d → i; B → c → iv;
C → b → iii; D → a → ii

(2) A → c → iii; B → b → iv;
C → d → ii; D → a → i

(3) A → b → i; B → c → iii;
C → a → iv; D → d → ii

(4) A → c → iii; B → b → i;
C → d → iv; D → a → ii

67) Fertilization in human take place in:

(1) Uterus

(2) Ovary

(3) Ampulla of fallopian tube

(4) Infundibulum

68) A structural and functional unit between developing embryo (foetus) and maternal body called–

(1) Umbilical cord

(2) Allantosis

(3) Yolk sac

(4) Placenta

69) Embryo with more than 8 blastomeres formed due to in vitro fertilisation is transferred into :

(1) Uterus

(2) Fallopian tube

(3) Fimbriae

(4) Cervix

70) Several hormones like hCG, hPL, estrogen, progesterone are production by

(1) Ovary

(2) Placenta

(3) Fallopian tube
(4) Glands

71) Heart sound of the foetus can be felt through the stethoscope as early as the end of

(1) 1st month
(2) 3st month
(3) 5st month
(4) 4st month

72) Medical termination of pregnancy is considered safe up to:

(1) 12 week
(2) 30 week
(3) 40 week
(4) 27 week

73) Which one of the following part is cut and tied in tubectomy :

(1) Uterus
(2) Fallopian tube
(3) Cervix
(4) Vagina

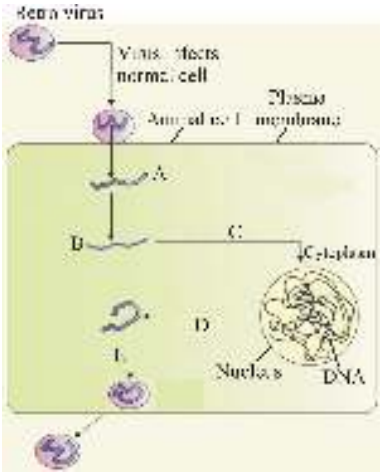
74) Late detection and improper treatment of STIs leads to:

(1) Pelvic inflammatory diseases
(2) Abortions, still birth, ectopic pregnancies
(3) Infertility or cancer of reproductive tract
(4) All of these

75) **Statement-I :-** Cu ions suppress sperms motility and fertilizing capacity of sperm.
Statement-II :- Hormone releasing IUDs in addition make the uterus suitable for implantation and the cervix hostile to the sperm.

(1) Both **Statement I** and **Statement II** are correct.
(2) **Statement I** correct and **Statement II** are incorrect.
(3) **Statement I** is correct but **Statement II** is incorrect.
(4) **Statement I** is incorrect but **Statement II** is correct.

76) The figure given below shows mode of action of AIDS virus. Identify steps A, B, C, D and E labelled in it.



(1) A-New viral DNA introduced into cell, B-Viral RNA produced, C-Viral DNA incorporated into host genome, D-New viral DNA, E-New viruses produced

(2) A-Viral DNA incorporated into host genome, B-Viral DNA, C-New viral RNA introduced, D-Viral RNA produced, E-New viruses produced

(3) A-Viral RNA introduced, B-Viral DNA, C-Viral DNA incorporated into host genome, D-New viral RNA produced, E-New viruses produced

(4) A-Viral DNA introduced, B-Viral RNA, C-Viral RNA incorporated into host genome, D-New viral DNA produced, E-New viruses produced

77) Which of the following statement is correct ?

(1) Atropa belladona and Datura shows hallucinogenic

(2) Cannabinoids act as degressant and slows sown body function.

(3) Cocaine is very effective sedative and painkiller

(4) Marphine is commonly called as smack.

78) Match the List-I and List-II.

	List-I		**List-II**
A	Bone marrow	i	Secondary lymphoid organ
B	Spleen	ii	A lobed organ
C	thymus	iii	Primary lymphoid organ
D	Acid in stomach	iv	Physiological barriea

Choose the correct answer from the options given below.

(1) A - ii, B - i, C - iii, D - iv

(2) A - iii, B - ii, C - i, D - iv

(3) A - iii, B - i, C - ii, D - iv

(4) A - i, B - ii, C - iv, D - iii

79) **Statement-I :** Nicotine stimulates adrenal gland to release adrenaline and non-adrenaline into blood circulation, both the which lower blood pressure and heart rate.
Statement-II : Smoking increases carbonmmoxide (CO) content in blood and reduce the concentration of haembound oxygen.

(1) Both **Statement I** and **Statement II** are correct.
(2) Both **Statement I** and **Statement II** are incorrect.
(3) Only **Statement I** is correct.
(4) Only **Statement II** is correct.

80) The process of spreading the cancerous cells to distant site is known as

(1) Hyperstasis
(2) Metastasis
(3) Parastasis
(4) Parasitesis

81) Antiserum is rich in:

(1) Steroids
(2) Antibodies
(3) Antigens
(4) RBC

82) X- rays are used in

(1) ECG
(2) EEG
(3) CT -Scan
(4) Endoscopy

83) Opioids are the drugs which bind to specific opioid receptors present in our CNS and GIT. Heroin, commonly called smack is chemically....., which is white, odourless, bitter crystalline compound, obtained by.....of morphine ?

(1) Diacetylmorphin, methylation
(2) Diacetylmorphin, acetylation
(3) Benzodiazepines, amination
(4) Amphetamines, acetylation

84) Which one of the following options gives correct categorization of examples and the two successive Phenomenon given in the columns?

	Examples	Phenomenon - I	Phenomenon - II
(1)	*Bision bataloria*	Industrial melanism	Artificial selection
(2)	Australian marsupials	Adaptive radiation	Convergent evolution
(3)	Placental mammals	Convergent evolution	Natural selection
(4)	Darwin's finch	Adaptive radiation	Divergent evolution

(1) 1

(2) 2

(3) 3

(4) 4

85) In 1938 a fish caught in South Africa happened to be a Coelacanth which was thought to be extinct. These animals called lobefins evolved into the first :-

(1) Amphibians

(2) Jawless fish

(3) Reptiles

(4) Birds

86) Match the Column I with Column II and choose the correct option

Column I		Column II	
(A)	Neanderthal man existed	(i)	18,000 years ago,
(B)	Agriculture started	(ii)	1,00,000-40,000 years back
(C)	Pre-historic cave art developed	(iii)	About 15 mya
(D)	Dryopithecus existed	(iv)	10,000 years back

(1) (A) - (i), (B) - (ii), (C) - (iii), (D) - (iv)

(2) (A) - (ii), (B) - (iii), (C) - (i), (D) - (iv)

(3) (A) - (ii), (B) - (iv), (C) - (i), (D) - (iii)

(4) (A) - (i), (B) - (iii), (C) - (iv), (D) - (ii)

87) Which of the following statement is correct ?

(1) The rate of disappearance of new forms is linked to the life cycle.

(2) Fitness is the end result of the ability to adapt and get selected by nature.

(3) Branching descent is the key concept of Lamarck.

(4) According to Darwin, evolution is driven by use and disuse of organs

88) The process by which new species arise from a single common ancestor is called :-

(1) Speciation

(2) Genetic drift

(3) Mutation

(4) Natural selection

89) Which type of selection is observed in moth (Biston betularia) during industrial melanism ?

(1) Stabilising

(2) Directional

(3) Disruptive

(4) Artificial

90) Which of the following statement is incorrect :-

(1) Excess use of herbicides, pesticides etc, has only resulted in selection of resistant varieties in a much lesser time scale.

(2) Lichens can be used as industrial pollution indicators.

(3) In stabilizing selection peak gets higher and narrower because more individual a carrier mean character value.

(4) Unit of natural selection is population.